STAR BLASTER

GAME DESIGN DOCUMENT

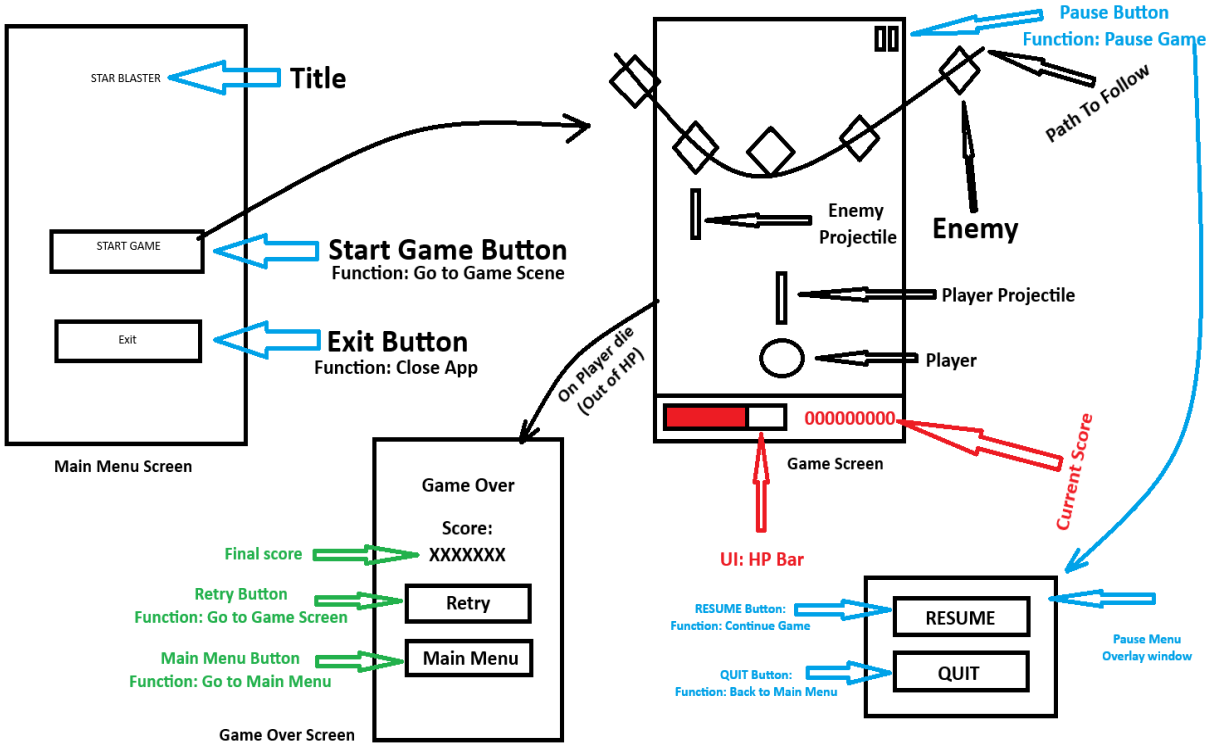
Genre: Shoot 'em up (STG) / Top-down Arcade Shooter

Platform: Mobile (Android/iOS) & PC / WebGL

1. **Core Loop & Game Flow**

+ Target: Players control a spaceship to attack, dodge incoming bullet hells from endless enemy waves, collect power-ups to enhance firepower, and achieve the highest score possible before being destroyed.

+ **Game flow:** Main Menu ➔ Gameplay (Combat / Dodge / Upgrade) ➔ Game Over & Score Summary ➔ Retry / Main Menu



GAME FLOW

2. **Core Mechanics & Control:**

Player Movement:

+ Controls: Supports both **Touch/Drag** (Mobile) and **Mouse Cursor** (PC/Editor). Ship position updates smoothly based on pointer position.

+ Screen Bounds: Movement constrained within the active Camera Viewport using Mathf.Clamp to prevent the player from moving off-screen or too far up. Shoot & Weapon:

Shooter & Weapon Systems:

+ Auto-fire Mechanism: Firing is continuously active upon holding touch or mouse press.

+ Cooldown & Anti-Spam: Fire rate is governed by an automated timer (nextFireTime / Coroutine) to enforce a strictly controlled attack frequency, neutralizing fast-click spamming.

Enemy Spawning & Behavior:

+ Pathing: Enemy waves spawn dynamically and follow pre-configured Waypoints (PathToFollow). The pathing system is modular, allowing easy addition, removal, or modification of trajectories.

+ Combat AI: Enemies fire projectiles based on assigned fire rates with randomized time variance (fireRateVariance) to create dynamic, organic bullet patterns.

+ Scoring: Points are awarded instantly upon defeating each enemy type.

3. **System & Economy**

Power-up System:

Power-ups spawn inside falling Space Asteroids after set intervals. Shooting and destroying an asteroid instantly grants a permanent or stackable buff to the player:

- 🔴 **Red Asteroid:** Attack Damage Boost (+Damage)
- 🟡 **Yellow Asteroid:** Fire Rate Boost (- Fire Rate)
- 🟢 **Green Asteroid:** Max HP Restore / Increase (+HP)

Dynamic Difficulty Scaling (DDS):

Game difficulty scales over time. Every fixed time interval (T_{scale}), enemy attributes (Max HP, Fire Rate, Movement Speed, and Damage) are multiplied by a growth factor to maintain high engagement and challenge.

4. User Interface

Main Menu Screen: Title, Start Game button (Scene transition), Exit button (ApplicationQuit).

Game HUD: Real-time Player Health Bar (HP), Current Score counter, and Pause Button.

Pause Overlay Window: Freezes time scale (Time.timeScale = 0), provides *Resume* and *Quit to Menu* actions.

Game Over Screen: Displays final calculated score, *Retry* button (Reload Gameplay Scene), and *Main Menu* button.